

Divyanshu Singh

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EDUCATION

Texas A&M University, College of Engineering, College Station

• **Master of Science, Computer Science**

May 2026

- **Research Focus:** Machine Learning–Powered Computer Vision for Human Modeling and 3D Reconstruction in Dynamic Scenes with Multiple Bodies

• **Bachelor of Science, Computer Science**

December 2024

- **Minor:** Mathematics

• **Courses:** Machine Learning, Artificial Intelligence, Operating Systems, Distributed Systems, Networks & Distributed Processing, Quantum Algorithms, Deep Reinforcement Learning, Deep Learning, Parallel Algorithms

SKILLS

Programming Languages: Java, Python, C/C++, MATLAB, SQL, HTML, CSS, JavaScript, Haskell, JSON

Certifications: AWS Cloud Practitioner

Frameworks: Flutter, Spring/Spring Boot, React.js, Express.js, Flask

Tools: GitHub, Linux/Unix, SSH, AWS, Google Cloud Platform

Other: Technical Writing, presentations, Microsoft Office

RELEVANT EXPERIENCE

Student Software Intern, Texas A&M University College Station, TX

Feb. 2024–Present

- Served as the primary developer of the [DataScribe.Cloud](#) data management platform, adopted by researchers across multiple universities and national laboratories.
- Contributed to the development of agentic AI components to support intelligent data workflows and collaborative research tasks.
- Designed and deployed AI-driven algorithms for data table schema recommendations, duplicate data detection, and automated data analysis.
- Developed a scalable microservices architecture and deployed containerized infrastructure using Docker and Kubernetes to streamline continuous integration and deployment pipelines.
- Authored a research paper on DataScribe platform, scheduled for publication in an academic journal.

Software Engineer Intern, Exabeam Inc. Foster City, California

May–Aug. 2023

- Implemented **Prometheus** and **Grafana**-based observability solutions, enabling real-time tracking of critical systems and reducing downtime
- Developed a **RESTful API** using **Spring Boot** for a cost analysis tool, allowing developers to assess the impact of code changes on job costs
- Deployed the tool to **Google Cloud Functions**, ensuring scalable execution and efficient resource utilization.
- Conducted security audits across GitHub repositories, identifying and remediating **CVEs** to enhance overall system security

Teacher Assistant, Texas A&M University College Station, TX

Spring 2024

- Created **Bash scripts** for automated testing and grading, improving efficiency in assignment evaluation
- Conducted lab sessions and office hours, enhancing student understanding of **C++** programming concepts
- Collaborated on developing instructional materials, contributing to improved course content and delivery

PROJECTS

Ipelint MVP 2: AI-Powered Patent Analysis (Python, TensorFlow/PyTorch, NLTK, Scikit-learn, AWS Bedrock)

- Developed **AI models** for Ipelint, a startup focused on patent prosecution analysis, emphasizing keyword/anti-keyword identification and diagram analysis
- Researched and implemented **fine-tuning** techniques, keyword extraction algorithms, and topic modeling to enhance model performance
- Developed **Computer Vision Model** for patent diagram analysis to enhance prior art search and collaborated with subject matter experts to gain domain knowledge in intellectual property and patent law

Camera View Selection for 3D Reconstruction using Deep Reinforcement Learning (Python, OpenAI Gym, COLMAP)

- Developed Deep Reinforcement Learning approach to intelligently select optimal camera views for multi-view 3D reconstruction.
- Implemented a custom Gym-compatible environment that simulates view selection and integrates with COLMAP for 3D reconstruction and quality metric calculation
- Trained DQN agent for sequential view selection, balancing reconstruction accuracy with computational efficiency

Parallel ICP Algorithm for Point Cloud Registration (C++, MPI, OpenMP)

- Implemented Iterative Closest Point (ICP) algorithm for 3D point cloud registration with MPI-based parallelization.
- Applied optimization strategies including dynamic resolution adjustment and KD-tree spatial indexing
- Achieved 76x speedup compared to base implementation through parallel processing optimization